Max Product Subset

Difficulty: **Medium** Accuracy: **17.21%** Submissions: **93K+** Points: **4**

Given an array **arr[],** find and return the maximum product possible with the **subset** of elements present in the array.

Note:

1. The maximum product can be of a single element also.
2. Since the product can be large, return it modulo $\mathbf{10^9 + 7}$.

Examples:

Input: arr[] = [-1, 0, -2, 4, 3]
Output: 24
Explanation: Maximum product will be (-1 * -2 * 4 * 3) = 24

Input: arr[] = [-1, 0]
Output: 0
Explanation: Maximum product will be (-1 * 0) = 0

Input: arr[] = [5]
Output: 5
Explanation: Maximum product will be 5.

Constraints:

$1 \leq arr.size() \leq 2 * 10^4$

$-10 \leq arr[i] \leq 10$

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class Solution:
    def findMaxProduct(self, arr):
        MOD = 10**9 + 7
        n = len(arr)

        if n == 1:
            return arr[0]

        prod = 1
        neg_count = 0
        zero_count = 0
        max_negative = -float('inf')

        for x in arr:
            if x == 0:
                zero_count += 1
                continue

            if x < 0:
                neg_count += 1
                max_negative = max(max_negative, x)

            prod *= x

        # All elements are zero
        if zero_count == n:
            return 0

        # One negative and rest zeros
        if neg_count == 1 and neg_count + zero_count == n:
            return 0

        # If count of negatives is odd, remove the negative
        # having the smallest absolute value
        if neg_count % 2 == 1:
            prod //= max_negative

        return prod % MOD
```

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